

Lecture on Deep Learning and Machine Learning Models

Neural networks are computing systems inspired by the biological neural networks that constitute animal brains.

We use backpropagation to calculate error gradients and update weights using gradient descent algorithms.

Linear regression models fit linear equations to predict continuous outcomes by minimizing mean squared error.

Logistic regression is used for classification tasks to predict class probabilities.

Overfitting occurs when a model fits training noise and exhibits high variance and low bias, resulting in poor validation accuracy.

We can apply dropout regularization to randomly deactivate a subset of neurons during training epochs.

Transformers have self-attention mechanisms that calculate query, key, and value dot products, allowing parallel sequence training.